

Eco Pulse

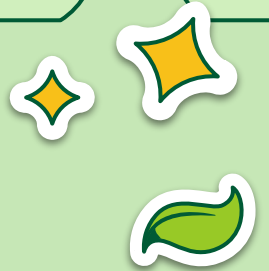
Waste Management Tracker

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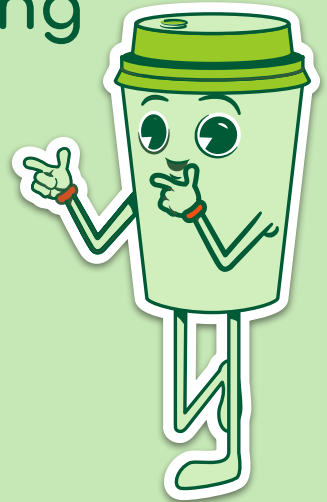




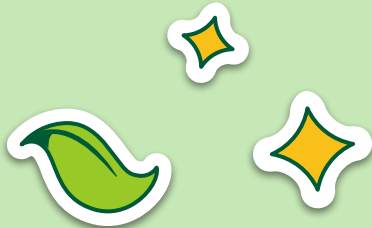
Problem Statement



Small Cafes across the US are losing money to avoidable waste.



What inspired me?



ASU Cafes

The Roast in Durham Hall and the Starbucks in the MU.

Tempe

Urban Boba and The Alley.

Opportunity to save money

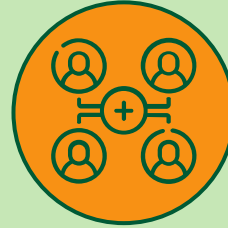
Even small reductions in waste can save money and improve sustainability

What did I use?



Streamlit

Creates the interactive web interface for logging and viewing the data.



Python

Mainly coded in python. Used it for all the logic and calculations.



Plotly

Helped me generate smooth dynamic graphs, trends, and data.



Pandas

Handles, cleans, and aggregates the waste CSV file.



What it does?

Logs daily waste by item

Organic, Paper, Glass, Metal

Display the cost of wasted items

Owners can enter in the cost associated with the waste item.

Creates visual Data

Creates a waste over time and cost over time charts and a category bar chart

Get reorder recommendations

Uses the data inputted in the waste log to make smart recommendations.

Highlights your top items

Displays the top 3 wasted items weekly.

Reset Waste log

Owners can reset the waste log.



Data Model

- Item Name
- Item Category (Plastic, Paper, Metal, Glass, Organic, Other)
- Quantity Wasted (kg)
- Reason
- Date
- Cost (\$)
- Location



EcoPulse

Enter the name of the waste item:

Croissant Dough

Select the category of the waste item:

Organic

Enter the quantity wasted (in kg):

6.00

Enter the reason for the waste:

Overstocked Dough



Enter the date of waste:

2025/09/30

Enter the cost associated with the waste (in \$):

36.00

Enter the location where the waste occurred:

Kitchen

Submit Waste Entry

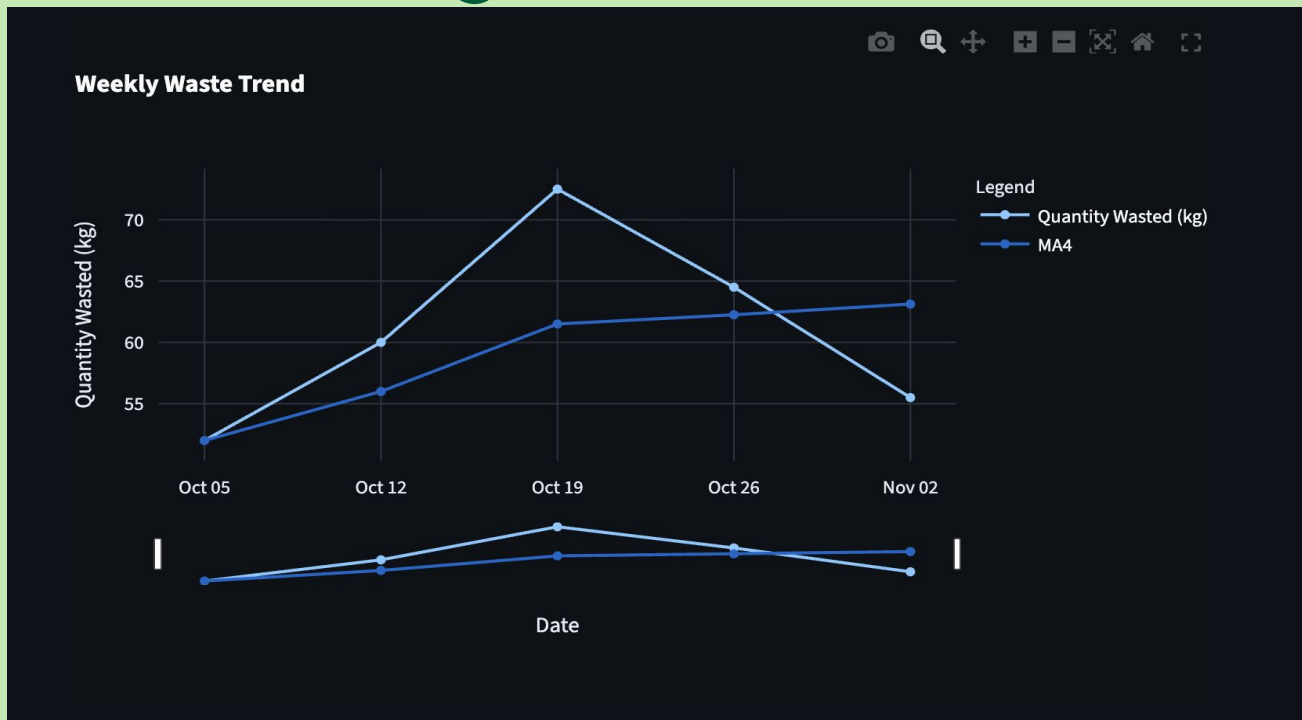


Insight Visuals



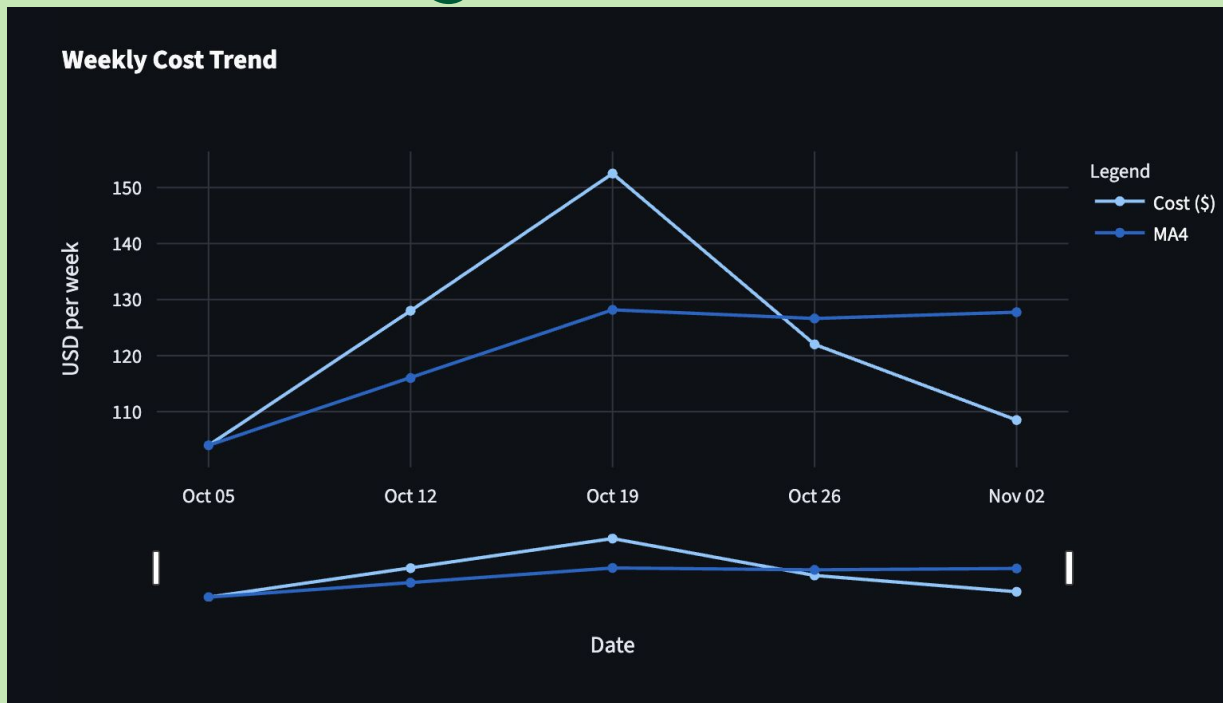


Insight Visuals





Insight Visuals



Recommendation Logic

Smart Recommendations

Automated Weekly Insights

Items with High Waste Share (>30%) in their Category this Week:

Order 20% less the next cycle for these items:

	Item Name	Item Category	ShareOfCategory
0	Coffee Cups	Paper	47.37
1	Croissant Dough	Organic	59.38
3	Paper Lids	Paper	34.21
4	Plastic Cups	Plastic	100
5	Sandwich Bread	Organic	40.62

Items in Top 3 Waste for 2 Consecutive Weeks

These items were in the Top 3 for 2 consecutive weeks:

	Item Name	Week 1	Week 2	Recommendation
0	Croissant Dough	2025-09-29/2025-10-05	2025-10-06/2025-10-12	Check the vendor/date rotation.
1	Sandwich Bread	2025-09-29/2025-10-05	2025-10-06/2025-10-12	Check the vendor/date rotation.
2	Coffee Cups	2025-09-29/2025-10-05	2025-10-06/2025-10-12	Check the vendor/date rotation.
3	Croissant Dough	2025-10-06/2025-10-12	2025-10-13/2025-10-19	Check the vendor/date rotation.
4	Coffee Cups	2025-10-06/2025-10-12	2025-10-13/2025-10-19	Check the vendor/date rotation.
5	Croissant Dough	2025-10-13/2025-10-19	2025-10-20/2025-10-26	Check the vendor/date rotation.

High Waste Share

It would display the items to order 20% less of.

Top 3 waste for 2 consecutive weeks

It would recommend to check the vendor/date rotation

Results from mock data



- Total Waste: 304.5 kg
- Waste log of over 30 items when from October 1st to November 2nd.

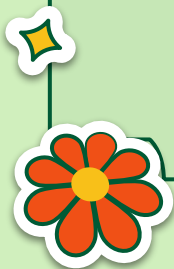
Waste Log

	Item Name	Item Category	Quantity Wasted (kg)	Reason	Date	Cost (\$)	Location
0	Croissant Dough	Organic	18.5	Overproofed	2025-10-01	45	Bakery Sta
1	Sandwich Bread	Organic	12	Stale	2025-10-01	24	Bakery Sta
2	Coffee Cups	Paper	8	Damaged	2025-10-01	12	Storage
3	Paper Lids	Paper	6	Torn	2025-10-01	8	Storage
4	Plastic Lids	Plastic	4.5	Broken	2025-10-01	10	Storage
5	Paper Straws	Paper	3	Wet	2025-10-01	5	Storage
6	Croissant Dough	Organic	20	Overproofed	2025-10-08	48	Bakery Sta
7	Sandwich Bread	Organic	15	Stale	2025-10-08	30	Bakery Sta
8	Coffee Cups	Paper	9.5	Damaged	2025-10-08	14	Storage
9	Paper Lids	Paper	7	Torn	2025-10-08	9	Storage

Total Waste: 304.5 kg

Top 3 Wasted Items

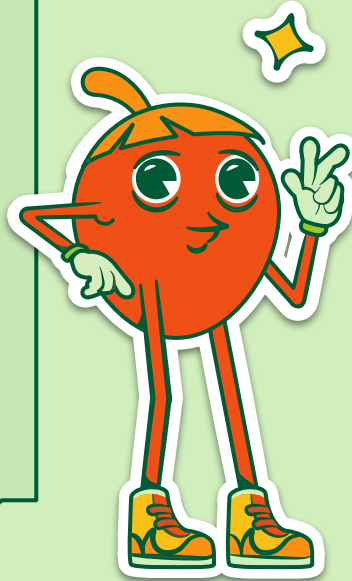
	Item Name	Quantity Wasted (kg)
3	Croissant Dough	114.5
11	Sandwich Bread	54
2	Coffee Cups	47.5



What's next?

Expand

Expanding the vendor/date rotation by implement weekly emails to the business owner so they are reminded.

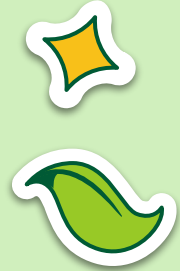
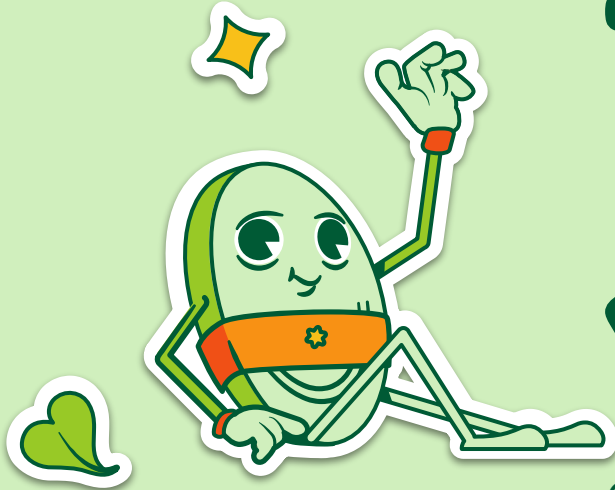


Export

Adding a feature to export the CSV file so it is saved for business owners for accounting purposes and for owners to do there own external analysis.



Thank you



Github: <https://github.com/Nelly444/Mentorship-Project>
Linkedin:
<https://www.linkedin.com/in/nelson-supriyasilp-1b9484322/>